Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/01/2009
PI Name:	Alexander, J. Iwan D. Ph.D.		
Project Title:	National Center For Space Exploration Research (NCSER)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Biofluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	05/28/2002	**End Date:**	06/30/2008
No. of Post Docs:	3	**No. of PhD Degrees:**	4
No. of PhD Candidates:	2	**No. of Master' Degrees:**	4
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received extension to 6/30/2008 per T. Rusnak/GRC (4/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3 975		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NCSER is a non-profit organization, created in 1997 as the National Center for Microgravity Research under the sponsorship of NASA through a cooperative agreement with Universities Space Research Association (USRA) and Case Western Reserve University. The NCSER is physically located on the campus of CWRU and at NASA GRC where NCSER scientists and engineers are work together with GRC personnel. The role of the NCSER is to • Perform research in fluids and combustion to support NASA's space program and related national initiatives • Increase awareness of microgravity research and enhance its scientific, technological, educational, and economic impacts • Support the development of enabling technologies for space exploration		

Task Description:

Originally chartered with supporting and developing NASA microgravity experiments on the 'shuttle' and ISS, the combined NCSER expertise in fluids and combustion processes in reduced gravity (i.e., microgravity, Lunar and Martian gravity) is second to none. Our current in-house expertise includes fluids management and control in reduced gravity, reduced gravity combustion processes; microfluidics, heat transfer and phase change processes; computational multiphase fluid dynamics and heat and mass transfer, computational simulation of biomedical processes in support of the 'Digital Astronaut' for autonomous space medicine. NCSER scientists and engineers are currently engaged in long term research related to NASA's exploration vision: Current investigations include: fire safety- flammability, suppression and detection, propellant management, and fluids processes for advanced human support technologies.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

NCSER was formed in February 2005, evolving from the former National Center for Microgravity Research (NCMR) in response to NASA's new research focus in its Vision for Space Exploration (VSE) and in order to better address topics of relevance. The NCMR was originally chartered with supporting NASA's microgravity research programs in fluids and combustion and facilitated fluids and combustion research by assembling a cadre of in-house subject matter expertise in these areas, and by serving as Technical Monitors for grants, Project Scientists, and project support staff for flight experiments. From 1997 to 2007, NCSER staff members were Principal Investigators (PIs) and Co-investigators (Co-Is) on numerous competitively awarded grants. An independent review panel convened by Dr. Eugene Trinh, then Chief of NASA's microgravity science division in 2002, stated that "NCMR is a vital and successful operation, effectively supporting NASA's program in many ways" and encouraged NASA to "regard NCMR as a kind of small-scale laboratory of innovative research management". Dr. Trinh concluded, "The Center has performed with distinction in its core responsibilities of supporting NASA's research programs in fluids and combustion".
More recently, as NASA has risen to the challenge of adjusting to its new role in space exploration, NCSER scientists and engineers have responded by successfully shifting focus and transitioning from activities conducted under NASA's former microgravity research program to critically new research and technology development relevant to space exploration. The flexibility and responsiveness demonstrated in this transition is due to the breadth of science and engineering disciplines resident within NSCER and their relevance to exploration systems technology developments at GRC and other NASA centers. Indeed, NCSER's past experience with the effects of the space environment, especially reduced gravity effects on multiphase fluid and reacting flow processes associated with exploration systems and technologies, is directly applicable to ongoing and anticipated exploration research and technology development and for defining and/or designing methods to mitigate the negative effects of the space environment.

During the period 2006-2008 amid NASA's restructuring of their programs at Glenn Research Center, the NCSER underwent a change in focus, moving away from a large role in supporting ISS microgravity research and toward research motivated by space exploration technology needs and challenges.. In particular, our staff scientists were able to apply their expertise in reduced-gravity fluids, combustion and reacting flow processes to research related to spacecraft thermal and fluid systems, fire safety and astronaut health. The NCSER focused around five activities during the period 2006-2008:

• Exploration Systems Technology Research

• Engineering Research Support

• ISS Research and Pl support

• Workshop and Conference Organization and Support

Because of its core competencies in the physical/chemical sciences and expertise in low-gravity fluids transport, combustion and reacting flow phenomena, the NCSER has established partnerships to deal with a broad spectrum of problems related to the development of enabling technologies for space exploration. These problems range from thermal control, water management and waste processing in life support systems, to fire safety, to multiphase fluid management in power and propulsion, in-situ resource utilization and on to bioscience and medical technologies and counter-measures. NCSER was operated through Cooperative Agreement NCC3-975 that continued through June 30th, 2008. The NCSER is a collaborative arrangement between the Universities Space Research Association (USRA), Case Western Reserve University and NASA Glenn Research Center (GRC). It combines the breadth and depth of USRA's nationally distributed organization with two of Northeast Ohio's research powerhouses to produce a potent combination of skills and expertise in reduced gravity fluids and combustion research that is applied to problems of significance. Thus, NCSER provides research avenues that, by filling gaps in knowledge, will lead to an important foundation upon which broadly applicable new technologies can be developed for use in future systems developed by various NASA Enterprises.

Bibliography

Hundreds of papers were published and presented by NCSER (NCMR) researchers and their collaborators. Selected examples are given below to illustrate the breadth and depth of the work that was reported.

Selected Refereed Publications arising from the cooperative agreement

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